

A smooth-symbol partial answer to the spectral open-disk question in arXiv:2005.09557

Literature-implied identification note

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Status: literature-implied answer (partial smooth-symbol subcase). No full answer is claimed for an arbitrary H^∞ analytic part.

Original question

Abakumov, Baranov, Charpentier, and Lishanskii consider on H^2 the Toeplitz operator with meromorphic symbol

$$\Phi(z) = R(1/z) + \varphi(z), \quad \varphi \in H^\infty,$$

where R has degree N and has no pole in the closed disk. They prove that hypercyclicity forces N -valence and record the spectrum formula

$$\sigma(T_\Phi) = \mathbb{C} \setminus \Phi(\mathbb{D}, N).$$

They then ask whether

$$T_\Phi \text{ hypercyclic} \implies \sigma(T_\Phi) \cap \mathbb{D} \neq \emptyset.$$

The question occurs on PDF page 3 of arXiv:2005.09557, immediately after the spectrum formula [1].

Supporting theorem and implication

On PDF page 14, Theorem 3.2 of Fricain, Grivaux, and Ostermann [2] says that if a Toeplitz symbol F satisfies

- (H1) $F \in C^{1+\varepsilon}(\mathbb{T})$ for a suitable $\varepsilon > 0$, and F' never vanishes;
- (H2) $F(\mathbb{T})$ has only finitely many simple self-intersections;
- (H3) $\text{wind}_F(\lambda) \leq 0$ off $F(\mathbb{T})$;

then hypercyclicity of T_F forces every connected component of $\text{int } \sigma(T_F)$ to meet the unit circle.

This gives a positive answer to the source question under (H1)–(H2). Indeed, the source's Proposition 1.1 says that hypercyclicity makes Φ N -valent. For $\lambda \notin \Phi(\mathbb{T})$, the argument principle gives

$$\text{wind}_\Phi(\lambda) = n_\Phi(\lambda) - N \leq 0,$$

so (H3) is automatic. Moreover, the regular immersed curve $\Phi(\mathbb{T})$ has a bounded complementary face adjacent along a regular arc to the unbounded face. The winding number is zero on the

unbounded face and changes by one across that arc; nonpositivity forces the bounded face to have winding -1 . It is therefore a nonempty component of $\text{int } \sigma(T_\Phi)$.

By Theorem 3.2 this open component contains a point of $\mathbb{T}$. Every neighborhood of a unit-circle point contains points of $\mathbb{D}$, hence

$$\sigma(T_\Phi) \cap \mathbb{D} \neq \emptyset.$$

Provenance and limitations

The supporting authors cite the original paper and devote a comparison section to its meromorphic-symbol setting, but the implication above is an agent-identified combination rather than a named answer in their paper. The unrestricted $\varphi \in H^\infty$ question is not resolved: the proof of the supporting theorem uses a bounded H^∞ functional calculus coming from smooth Toeplitz model theory. Radial smoothing is insufficient because hypercyclicity is not stable under strong or norm perturbation.

Bounded searches used the exact source sentence, both arXiv identifiers, the paper titles, and variants of “hypercyclic Toeplitz spectrum open unit disk.” No full later answer was found.

References

- [1] E. Abakumov, A. Baranov, S. Charpentier, and A. Lishanskii, *New classes of hypercyclic Toeplitz operators*, arXiv:2005.09557 (2020; revised 2021).
- [2] E. Fricain, S. Grivaux, and M. Ostermann, *Hypercyclicity of Toeplitz operators with smooth symbols*, arXiv:2502.03303, Theorem 3.2.